# MILITARY SPECIFICATION
# POLYETHYLENE
# (FOR USE IN AMMUNITION CONTAINERS)

## 1. SCOPE

1.1 This specification covers molded polyethylene for use, as filler pads in ammunition containers.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781 - Steel Strapping, Flat
PPB-B-1055 - Barrier Material, Waterproofed, Flexible

MILITARY

MIL-A-2550 - Ammunition and Special Weapons, General Specification for

STANDARDS

FEDERAL

FED-STD-595 - Colors

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105).
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-731 - Quality of Wood Members for Containers and Pallets
MIL-STD-1169- Packaging, Packing and Marking for Shipment of Inert Ammunition Components

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply:

American Society for Testing and Materials Publication

D1238-65T - Measuring Flow Rate of Thermoplastics by Extrusion Plastometer

D1505-67T - Measurement of Density of Plastics by the Density Gradient Technique

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

3. REQUIREMENTS

3.1 Material.-The material used in the manufacture of the polyethylene sheets shall be virgin or re-processed polyethylene meeting the requirements specified herein.

3.2 Fracture.-The polyethylene sheets shall be capable of bending without cracking or delamination when tested as specified in paragraph 4.4.1.

3.3 Physical property values.-The physical property values of the polyethylene sheets shall meet the applicable requirements of Table I when tested in accordance with the test procedures of paragraph 4.4 of this document.

TABLE I

Physical Property Requirements

| Property | Requirement |
|---|---|
| Density, gms/ml. | 0.916 Minimum (Min.) |
| Melt Index, grams/10 minutes with 2160 g load (see 6.5) | 5.0 minimum |
| Color Number (FED-STD-595) | The polyethylene sheets shall closely approximate any of the following composition color numbers. |



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| | |
|---|---|
| 14036 | 16165 |
| 14052 | 16187 |
| 14056 | 16251 |
| 14064 | 16307 |
| 14077 | 16314 |
| 16081 | 16357 |
| 16099 | |

Although specified colors are listed as Gloss, the surface finish on the polyethylene sheets will be as produced from good molding practices.

3.4 Thickness.-The polyethylene sheet shall have a thickness of 1/16 inch plus or minus (±) 5 percent.

3.5 Size.-Unless otherwise specified in the contract or order, the polyethylene sheets shall be furnished in sheets 4 feet by 8 feet in size.

3.6 Stiffness.-The polyethylene sheets shall not exceed the permissible drape of 40 degrees maximum (max.) when tested as specified in 4.4.4.

3.7 Workmanship.-Workmanship shall be in accordance with the best commercial practice of the industry. Finished sheets shall be free of metal inclusions and any odor not normally associated with the chemical composition of polyethylene or its production. Scrap polyethylene material may be reground and mixed, regardless of color, provided the end product meets the requirements of this document.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

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4.1.1 Submission of product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.
b. Specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Inspection provisions.

4.2.1 Inspection provisions.-The polyethylene sheets shall be visually or dimensionally inspected, as applicable, for compliance with the dimensional requirements and for workmanship. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.2.2 Packing and marking.-Packing and marking shall comply with all requirements specified within this specification.

4.3 Testing.

4.3.1 Sampling.-The sampling plans for each of the following tests shall be in accordance with Standard MIL-STD-105, Table IIA using Code Letter D and an AQL of 0.40 for major defects and an AQL of 0.65 for each minor defect. Samples shall be selected at random from 5,000 finished sheets.

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| Defect | Category | Method of Inspection | Code No. |
|---|---|---|---|
| Fracture (see 3.2) | Major defect | Test | 01001 |
| Stiffness (see 3.6) | Major defect | Test | 01002 |
| Thickness (see 3.4) | Minor defect | Gage | 01003 |
| Size (see 3.5) | Minor defect | Scale | 01004 |
| Color (see 3.3) | Minor defect | Visual | 01005 |
| Workmanship (see 3.7) | Minor defect | Visual | 01006 |

4.3.2 Density (see 3.3).-Major defect, Code No. 02001.-Three determinations shall be made on samples selected from 5,000 finished sheets. Failure of any value to meet the specified requirement when tested as specified in 4.4.2 shall be cause for rejection of the lot.

4.3.3 Melt index (see 3.3) Minor defect, Code No. 02002.-Three determinations shall be made on samples selected from 5,000 finished sheets. Failure of any value to meet the specified requirement when tested as specified in 4.4.3 shall be cause for rejection of the lot (see 6.5).

4.3.4 Inspection equipment.-For the performance of all tests and examinations specified in 4.3 and 4.4, commercial inspection equipment should be employed (see 6.6). The contractor shall have available, and utilize correctly, this equipment and is charged with the responsibility of insuring that proper calibration procedures are followed. Government approval of all inspection equipment is required prior to its use for acceptance purposes.

4.4 Test Methods and Procedures.

4.4.1 Fracture.-Upon conditioning at ambient temperature, the polyethylene sheet shall be manually bent around a mandrel 180 degrees with a radius equal to 96 times the thickness of the specimen. Any evidence of cracking or delamination shall be cause for the lot to be rejected.

4.4.2 Density.-The density of the specimen shall be determined in accordance with ASTM D1505-67T. The test specimens shall be cut from three individual sheets. Specimens shall not be annealed or otherwise treated so as to change the crystallinity (see 6.4).



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4.4.3 Melt Index.-Melt index shall be determined in accordance with ASTM D1238-65T. The test specimens shall be cut from individual sheets.

4.4.4 Stiffness.-At ambient temperature the polyethylene sheet shall be positioned on a table allowing 12 inches to extend therefrom. A load, sufficient in weight to prevent movement of the sheet, shall be applied at the base of the extended position (see Figure 3). The angle of drape shall be measured at both ends of the extended portion using standard measuring equipment. The lot will be considered rejected if either of the two sides of the sheet (draped portion) exceeds the specified requirement.

## 5. PREPARATION FOR DELIVERY

5.1 Packing.-Packing shall be Level A, B, or C, as specified.

5.1.1 Level A.

5.1.1.1.-Polyethylene sheets shall be packed on a pallet constructed as specified in 5.1.1.1.1. The unit load of polyethylene sheets shall be built up within the following limitations:

a. Overall height of the pallet and stack of polyethylene sheets shall not exceed 54 inches.

b. Net weight shall not exceed 2000 pounds.

5.1.1.1.1 Construction of pallet.-The pallet shall have an overall length and width dimension 1 1/2 inches greater than the dimensions of the polyethylene sheets being packed. The pallet shall be constructed of soft wood No. 2 common or equal. The pallet shall consist of load bearing top deck boards, stringer boards, posts and bottom deck boards. The quality, with respect to moisture content, defects, etc., shall conform to Class 2 of MIL-STD-731. All boards shall be surfaced top and bottom to a uniform thickness. The dimensions of the pallet components and construction shall be in accordance with Figure 1. The posts shall be surfaced on two sides to provide a uniform thickness between the lower deck boards and the stringers. The row of intermediate posts in the longer direction of the pallet shall be equidistant one from the other on not more than 25 inches centers. In the shorter dimension, there shall be three rows of posts with the center row of posts positioned equidistant from the outside row of posts. Nails used in the pallet construction shall be either flathead steel, cement-coated, or twist shank drive with

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spiral flutes having a minimum of four (4) flutes per mail (helical angle of flute to nail axis shall be approximately 30 degrees). Each bottom deck board shall be nailed to each pallet post with not less than three 6-penny or two (2) inches by 0.135 inch diameter twist drive nails. Each top deck board joining stringer boards and pallet posts shall be nailed, at each joint, with three (3) 12-penny or 3 1/4 inches by 0.135 inch diameter twist drive nails, except when board covers strap slot. Nailing of top deck boards to stringer boards at other than post locations shall require a minimum of three clinched common nails not less than two (2) inches in length at each joint, except where a board covers a strap slot. Where a board covers a strap slot, only two nails shall be used, one on each side of the slot at each joint.

5.1.1.1.2 Method of placing, shrouding and strapping.-The stack of polyethylene sheets shall be positioned on the pallet allowing the pallet 3/4 inch beyond the load on all four sides. A waterproof shroud shall cover the stacked polyethylene sheets on top, both sides and both ends. The shroud shall meet the requirements of Class H-1 material of Specification PPB-B-1055. The shrouded load shall be secured to the pallet with steel strapping conforming to QQ-S-781, Type I, Class A or B; size not less than 5/8 inch by 0.023 inch or 3/4 inch by 0.020 inch, properly tensioned and sealed. A piece of lumber of the same material as the pallet deck boards, nominal one (1) inch thick by four (4) inches wide and of a length equal to the height of the load of polyethylene sheets, shall be positioned under each strap on all sides of the load to afford protection to the sheets from the tensioned steel straps. The straps shall be centered on the lumber and secured with staples spaced not more than 12 inches apart nor more than 12 inches from the edge of the load. Figure 1 illustrates a Level B pallet load of polyethylene sheets size 96 inches by 48 inches, without a shroud. Not less than three (3) straps shall be used to encircle the load girthwise for a load of sheets up to 96 inches in length. Two straps shall be near the ends of the load and positioned under the second top deck board from each end. Intermediate strap shall be spaced approximately midway between end straps and positioned under top deck boards. Strapping grooves shall be provided in the stringer boards (see Figure 1). Three straps shall encircle the load lengthwise; two straps shall be positioned adjacent to the inside surfaces of the outside posts and one strap shall be positioned adjacent to and at either side of the center posts.

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5.1.2 Level B.

5.1.2.1 The polyethylene sheets shall be packed in conformance with 5.1.1 except that shrouding is not required.

5.1.3 Level C

5.1.3.1 The polyethylene sheets shall be packed in conformance with 5.1.2 unless otherwise specified in the contract or order (see 6.2).

5.2 Marking

5.2.1 Marking shall be in accordance with MIL-STD-1169.

6. NOTES

6.1 Intended use.-Polyethylene sheets are designed for cutting into filler pads for use where water-resistant material is required and where resistance to dimensional change under compression rather than resiliency is important.

6.2 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Level of pack (Level C - Type of pack.)
c. Size of sheets (if other than 4 feet x 8 feet).

6.3 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.4 Density specimens.-Saw speed in excess of 500 to 600 lineal feet per minute have been found to be unsatifactory as excess speed can essentially melt the material and greatly change the density.

6.5 Melt Index.-Polyethylene sheets with a melt index lower than that specified will be considered acceptable provided all other requirements of this document are met.

6.6 Inspection equipment.

6.6.1 Commercial inspection equipment is defined in AMC Regulation 702-2.



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6.6.2 Contractor-designed equipment.-In the event that a contractor elects to design his own inspection equipment, details of the designs (drawings, description, materials, etc.) shall be submitted to: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND12, Dover, New Jersey 07801, for approval prior to fabrication and use. Approval of such designs may be delegated to the contract administration office for minor defects only.

Custodian:
Army-MU

Preparing Activity:
Army-MU

Project Number: 9330-A453

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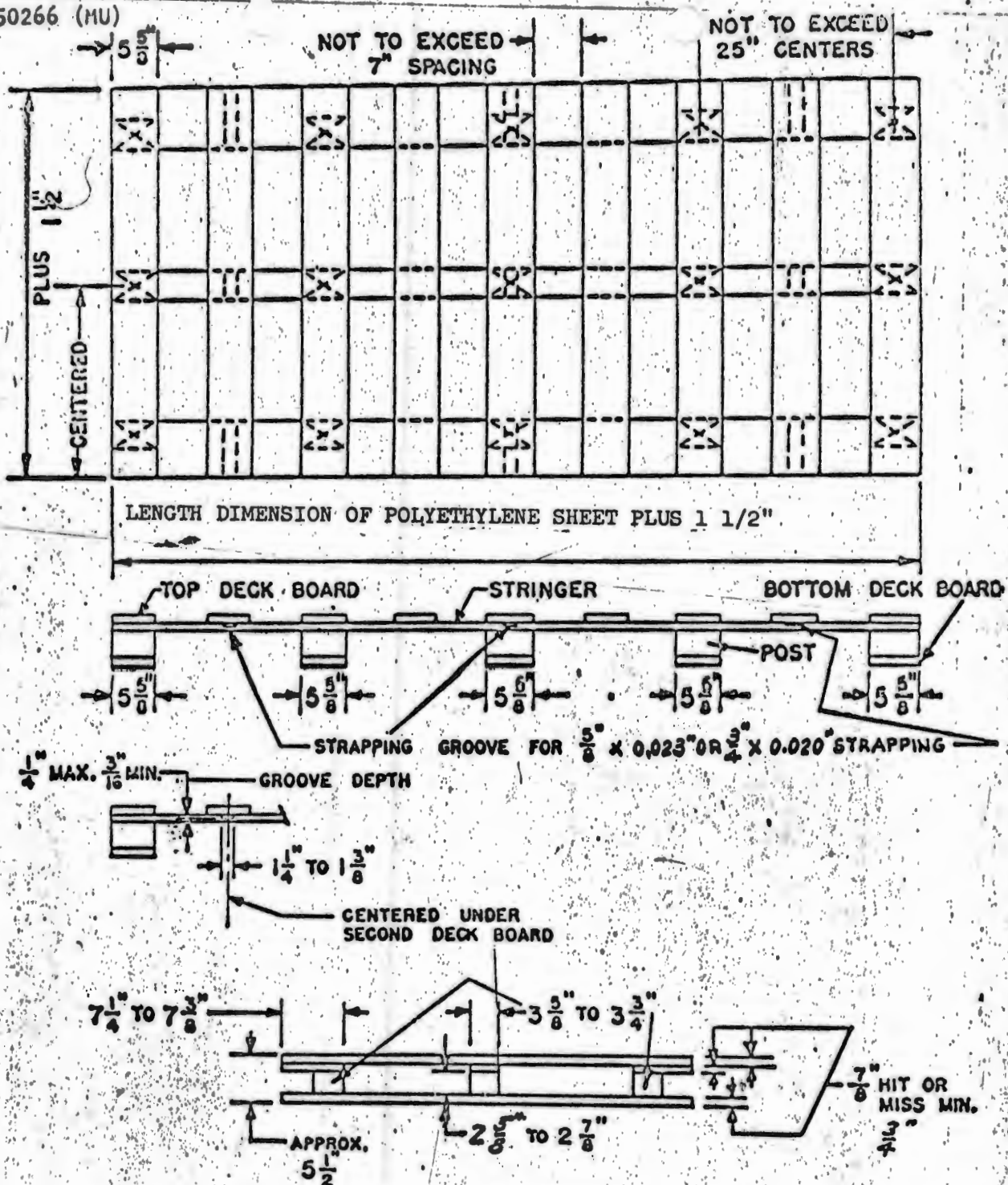



FIGURE 1. *Pallet construction.*

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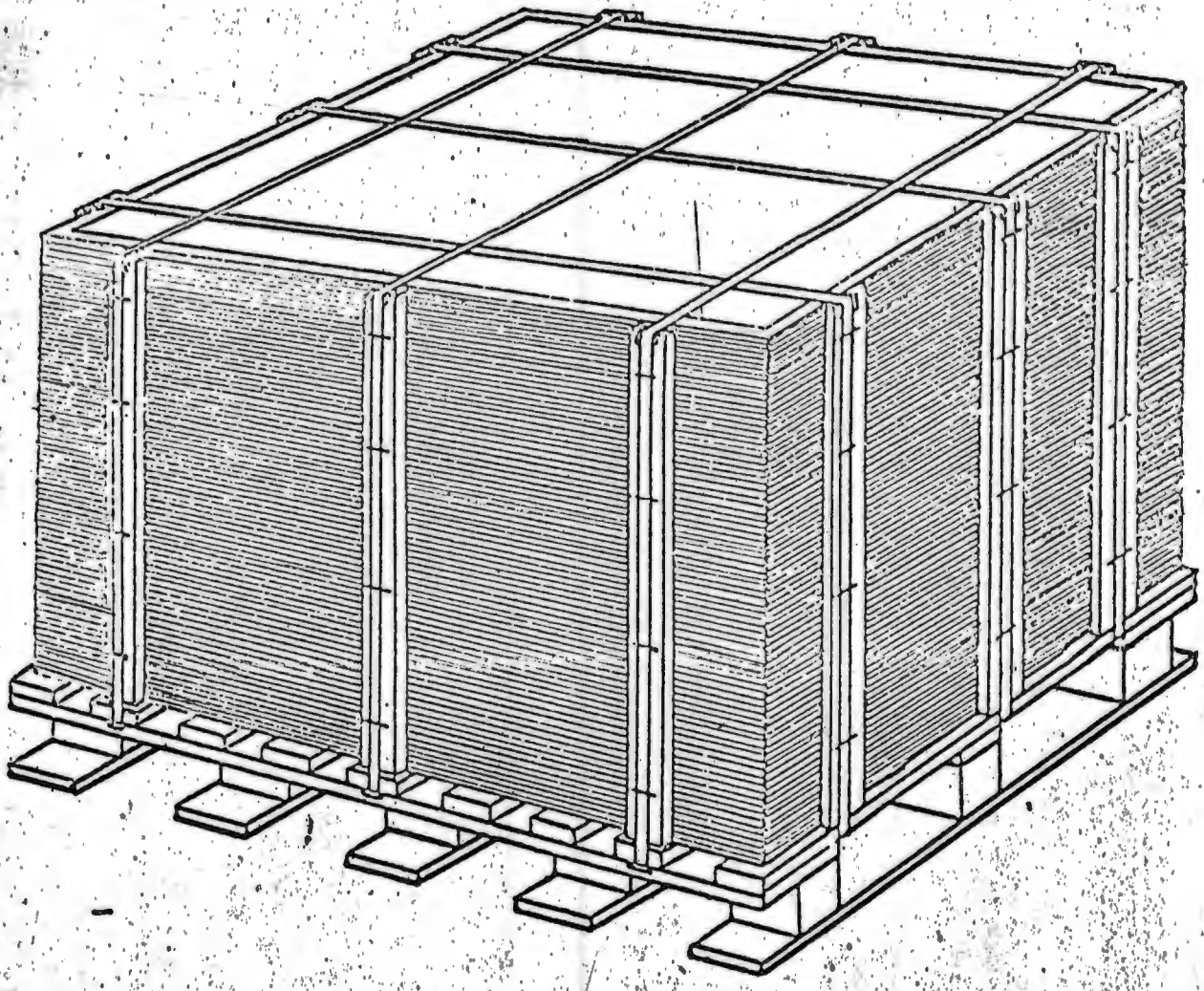


Figure 2.-Pallet load of polyethylene sheets, size 96" by 48"

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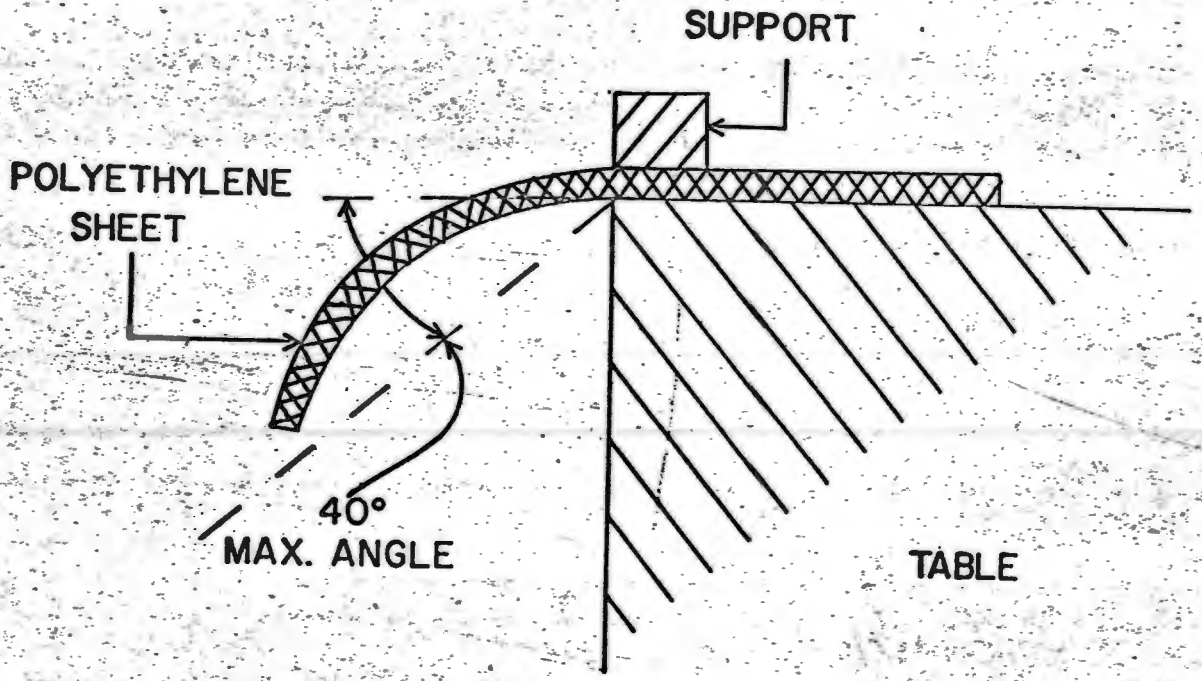



FIGURE 3



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